LUXURYSTAY

Hotel Management System

eProject Documentation

MERN Stack Web Application

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Appendix A. Interface Screens

This report is intentionally structured as a clean presentation document: each topic appears once, screenshots are grouped with the module they explain, and repeated branding has been removed.

1. Executive Summary

LuxuryStay Hotel Management System is a role-based web application developed to manage the operational lifecycle of a hotel from one connected platform. The system combines a public hotel website with authenticated Guest, Administrator and Manager workspaces.

The application supports room inventory, online reservation requests, booking confirmation, guest verification, check-in and check-out, housekeeping, maintenance, finance, reporting and feedback. Data is stored in MongoDB and accessed through an Express/Node.js backend, while the user interface is implemented in React.

Project Outcome

- Centralized hotel operations with separate role-based interfaces.
- Real-time room and booking status reflected across operational modules.
- Guest reservation workflow with identity and stay information.
- Administrative control for rooms, bookings, staff, finance, reports and feedback.
- Manager workflow for arrivals, departures, room status, housekeeping, maintenance and guest records.
- Persistent MongoDB storage for core hotel data.

2. Project Overview

2.1 Background

Hotel operations involve multiple connected activities: room availability, reservations, guest records, payments, arrivals, departures, cleaning, maintenance and management reporting. Handling these activities in disconnected files or manual registers can cause duplicate records, delayed updates and inconsistent room status.

2.2 Problem Statement

The project addresses the need for a single digital system where guests can request stays and hotel personnel can process the complete operational cycle without switching between unrelated tools.

2.3 Proposed Solution

A responsive MERN-stack hotel management system was created with a public-facing website and three controlled work areas: Guest, Administrator and Manager. Each role receives only the functions relevant to its responsibilities.

2.4 Primary Users

Role	Purpose
Guest	Search rooms, request reservations, review bookings and manage profile.
Administrator	Control rooms, reservations, staff, finance, reports and feedback.
Manager	Operate arrivals, departures, room status, housekeeping, maintenance and guest directory.

3. Objectives and Scope

3.1 Objectives

- Provide a clear online room and reservation experience.
- Maintain accurate room inventory and operational status.
- Create a controlled reservation approval and verification process.
- Support complete check-in and check-out workflows.
- Automatically connect departure operations with housekeeping preparation.
- Record maintenance activity and completed service history.
- Provide financial summaries, billing status and management analytics.
- Collect guest feedback after completed stays.
- Apply role-based authentication for protected areas.

3.2 Implemented Scope

The implemented scope covers the complete working flow demonstrated in the application: public browsing, authentication, guest booking, administrative confirmation, manager check-in/check-out, housekeeping, maintenance, finance, reporting, feedback and MongoDB persistence.

3.3 Scope Boundaries

Advanced capabilities such as automated email/SMS notifications, demand forecasting, dynamic pricing, enterprise backup automation and a full external payment gateway are not presented as implemented features. They are listed later as future enhancements.

4. Requirements and Technology Stack

4.1 Functional Requirements

Area	Implemented Function
Authentication	Register, sign in, role-based redirection and protected access.
Rooms	Create, view and manage room inventory and operational status.
Bookings	Request, confirm/reject, verify and track reservation states.
Guest Operations	Profile, bookings, identity/stay details and feedback.
Manager Operations	Check-in, check-out, room status, housekeeping, maintenance and guests.
Finance	Collected, outstanding and folio/payment status summaries.
Reports	Bookings, occupancy, revenue, room performance and booking activity.
Database	Persistent collections for users, rooms, bookings, contacts, feedback and maintenance.

4.2 Technology Stack

Layer	Technology
Frontend	React + Vite, React Router, Framer Motion, Lucide React
Backend	Node.js, Express.js
Database	MongoDB with Mongoose
Authentication	JWT-based authenticated sessions and role checks
Development	VS Code, npm, Git, MongoDB Compass
Environment	Windows 10; local development server and database

5. System Architecture and Workflow

5.1 Architecture

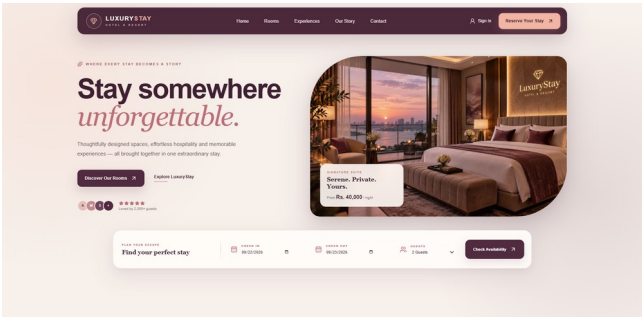
Layer	Responsibility
React Client	User interface, navigation, forms, dashboards and API requests.
Express API	Authentication, validation, business logic and route handling.
Mongoose Models	Application data structure and database access.
MongoDB	Persistent hotel records.

5.2 Core Reservation Workflow

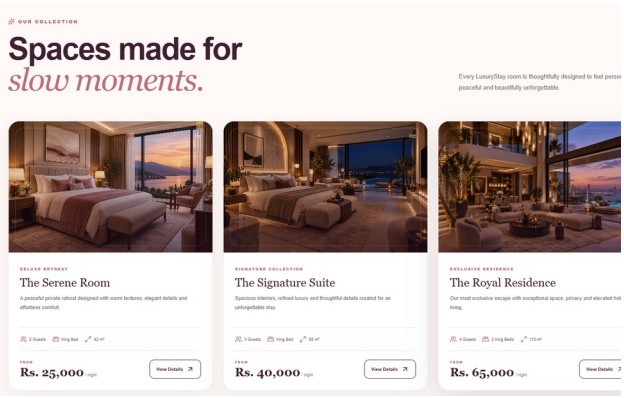
1. Guest selects a room, stay dates and guest count.
2. Guest provides contact, identity and reservation information.
3. Reservation is stored with a pending/verification state.
4. Administrator reviews the booking and confirms or rejects it.
5. Confirmed reservation becomes available to the Manager arrival queue.
6. Manager completes guest check-in and room status becomes occupied.
7. Manager completes check-out; the room can enter housekeeping preparation.
8. Housekeeping completes service and the room returns to available status.
9. Completed stays can be reviewed through feedback, finance and reporting modules.

6. Public Website and Authentication

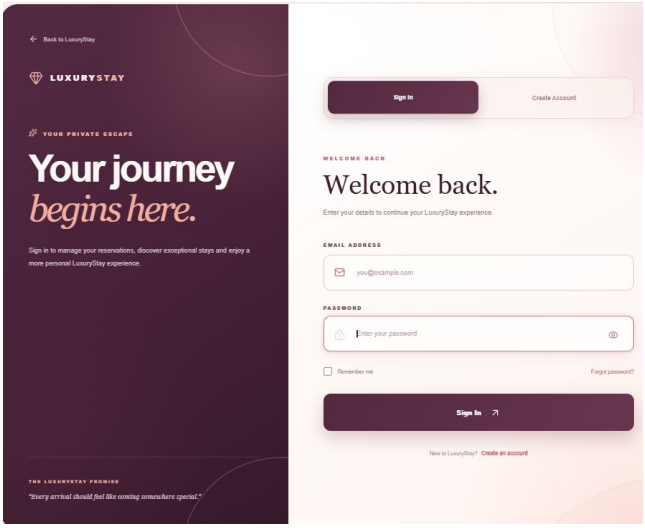
The public interface introduces the property, rooms, experiences and contact options. Authentication separates public browsing from protected hotel functions.



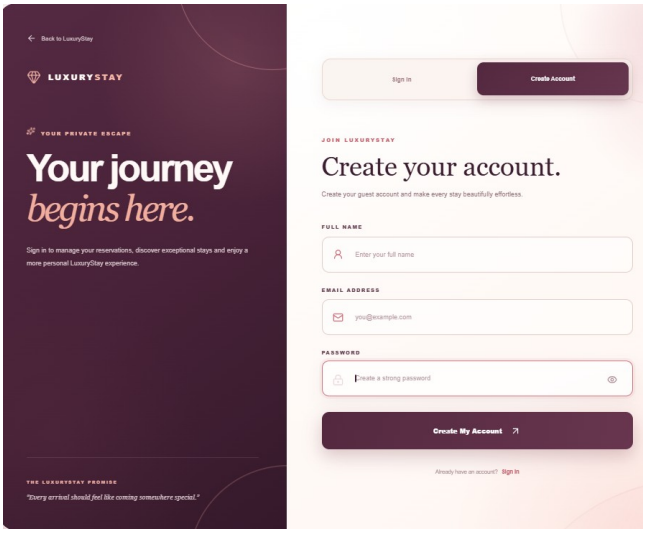
Homepage and booking search



Room collection and nightly pricing



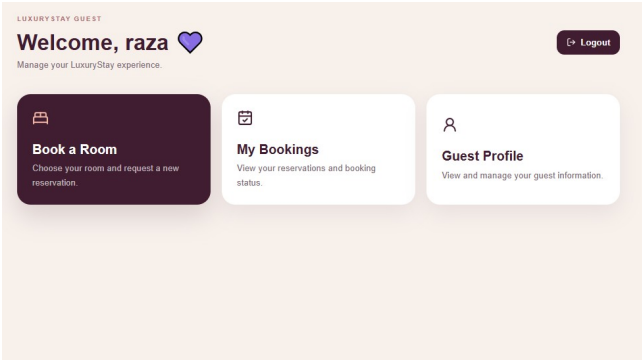
Sign In



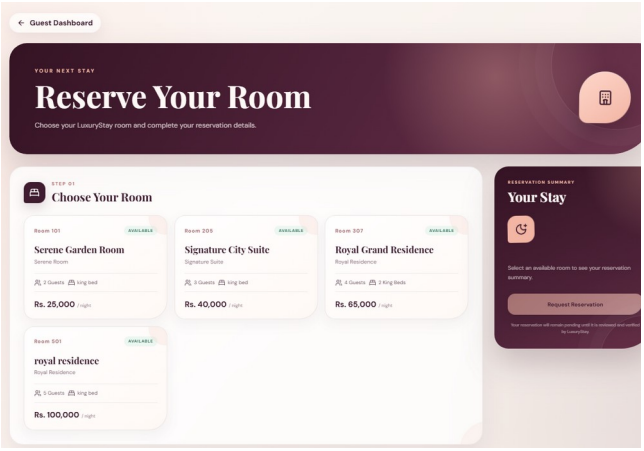
Create Account

7. Guest Module

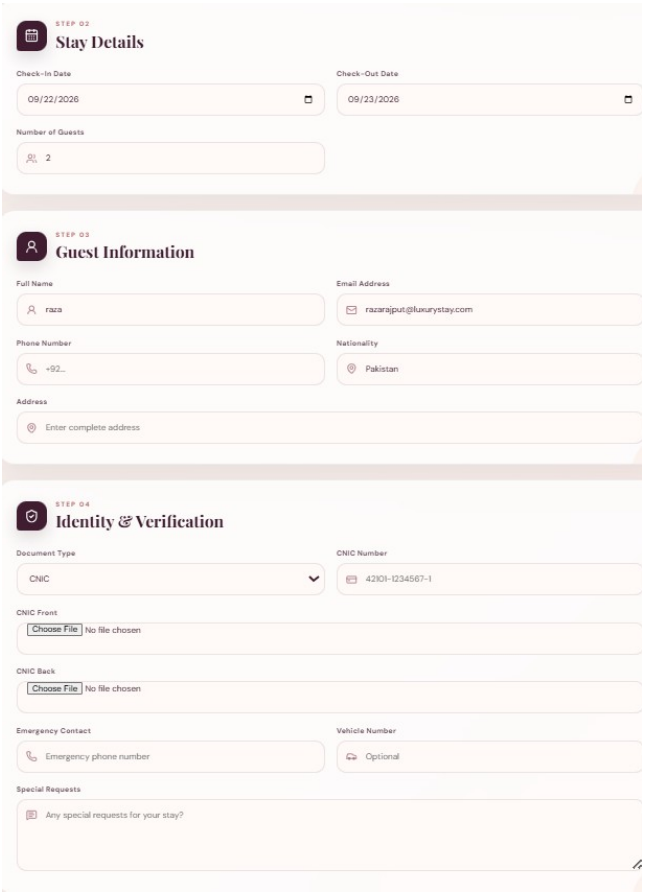
The Guest workspace provides direct access to booking, reservation history and profile management.



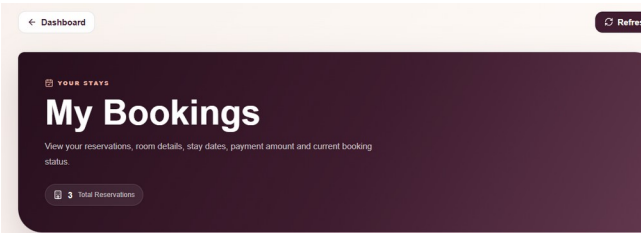
Guest Dashboard



Book Room - room selection

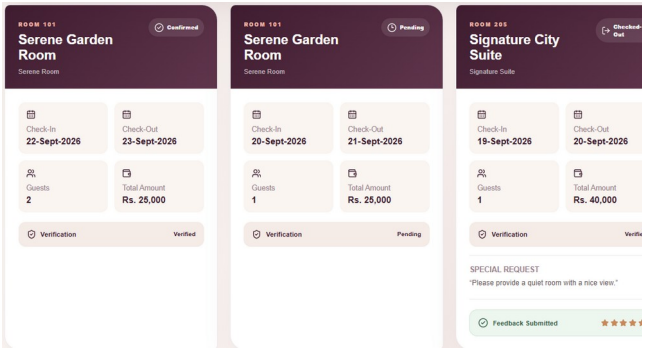


Book Room - stay and verification details

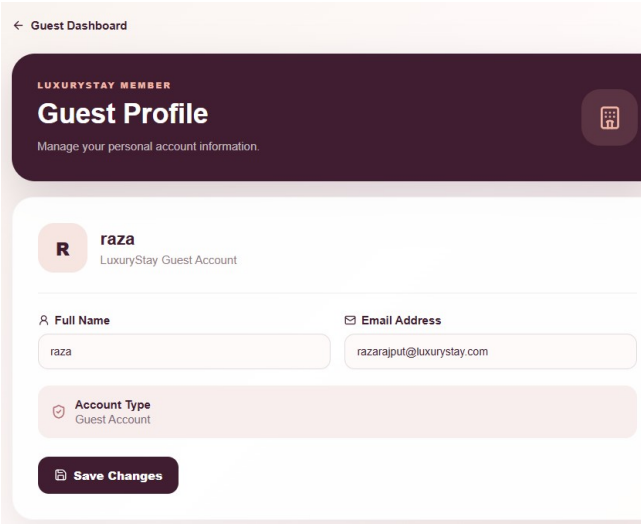


My Bookings summary

7.1 Booking and Profile Experience



Reservation states and stay information

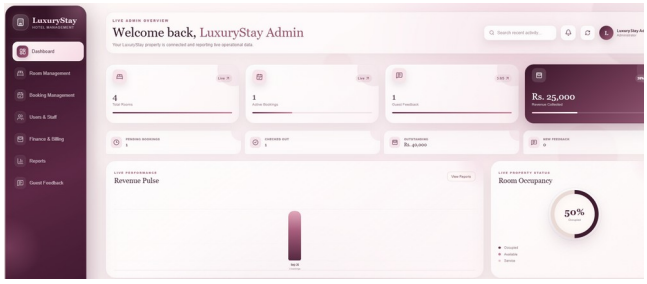


Guest Profile

- Available rooms are displayed with room type, capacity and nightly price.
- Stay dates and guest information are captured before reservation submission.
- Booking history displays status, dates, amount and verification state.
- Profile information can be reviewed and updated from the Guest workspace.

8. Administration Module

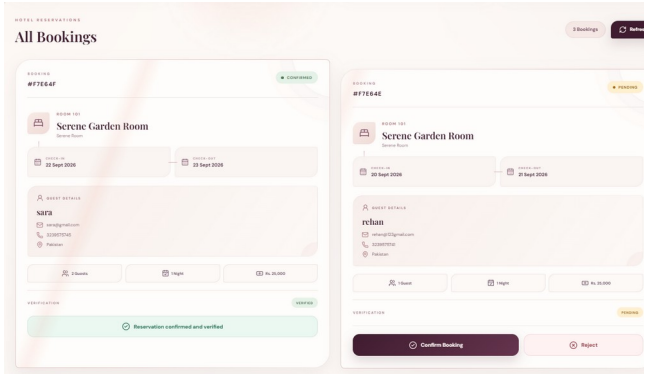
The Administrator workspace is the control center for property configuration and reservation administration.



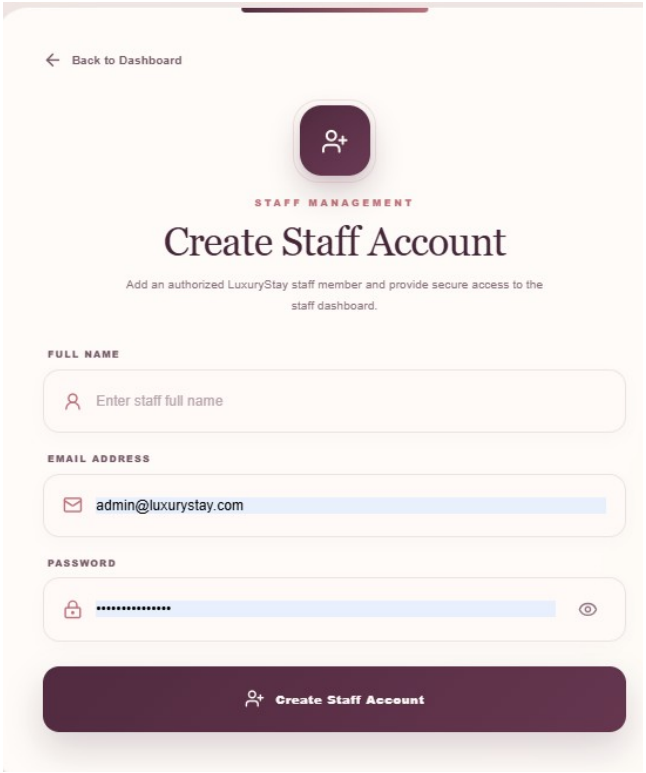
Admin Dashboard overview



Room inventory



Booking review and actions



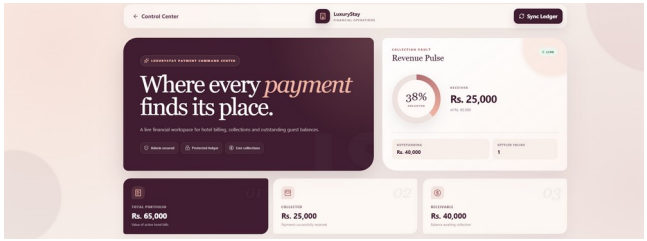
Create Staff Account

Administrative Responsibilities

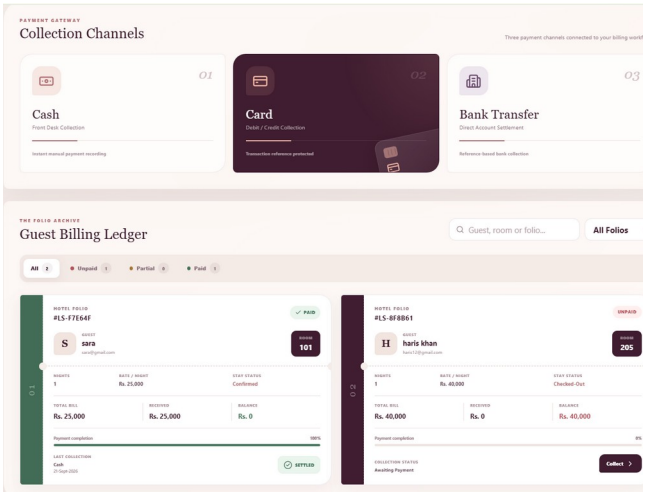
- Monitor operational KPIs and recent activity.
- Create and manage room inventory.
- Review pending reservations and verify booking information.
- Create authorized staff/manager accounts.
- Access finance, reporting and guest feedback modules.

9. Finance, Reports and Feedback

9.1 Finance



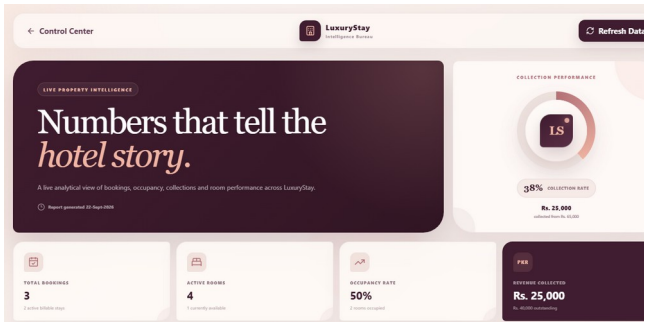
Finance overview and revenue pulse



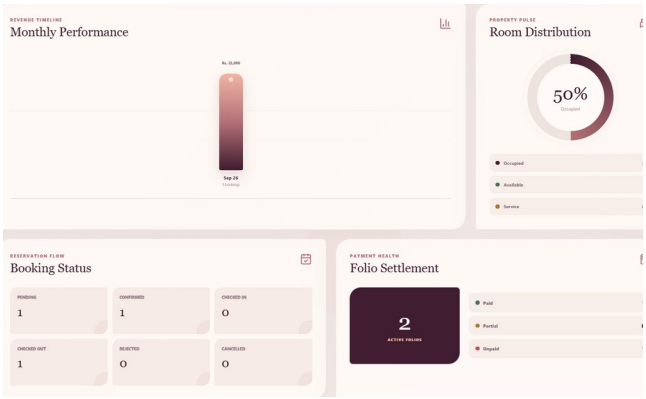
Billing ledger and collection channels

The finance workspace summarizes total portfolio value, collected revenue, receivables and folio settlement status. Cash, card and bank transfer are represented as collection channels.

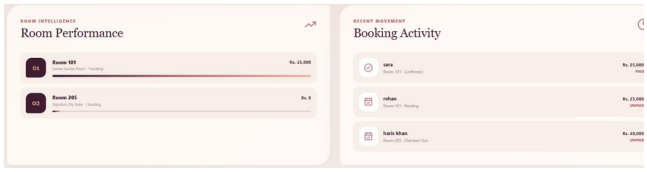
9.2 Reports and Analytics



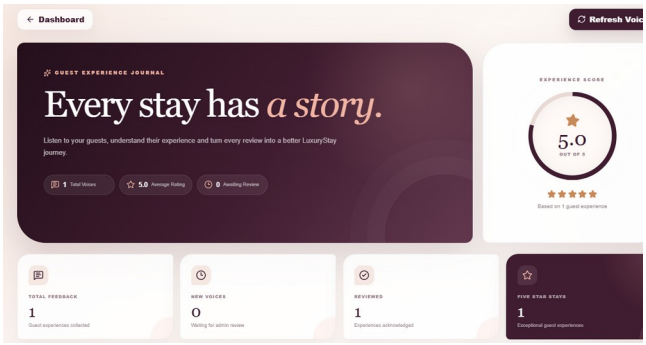
Reports overview



Monthly performance and booking status



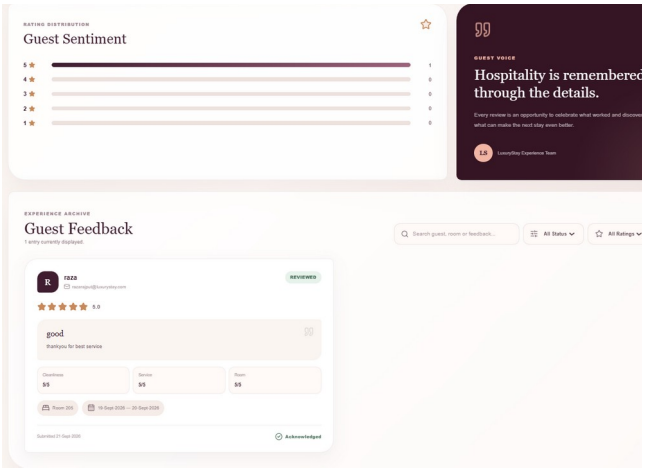
Room performance and booking activity



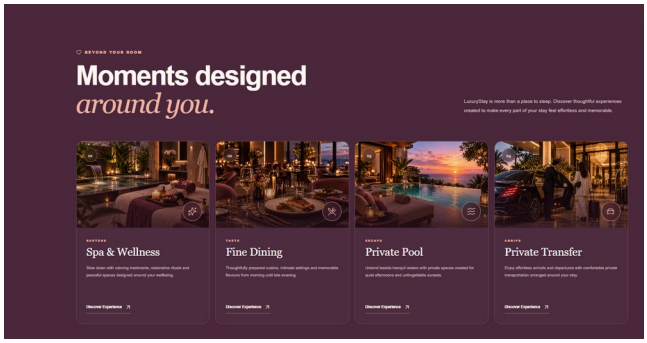
Guest experience score

Reports combine booking, room, occupancy and revenue information into management-focused visual summaries.

9.3 Guest Feedback



Guest sentiment and detailed feedback

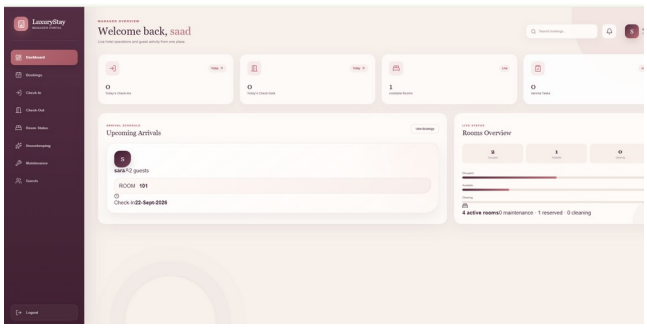


Public experiences section

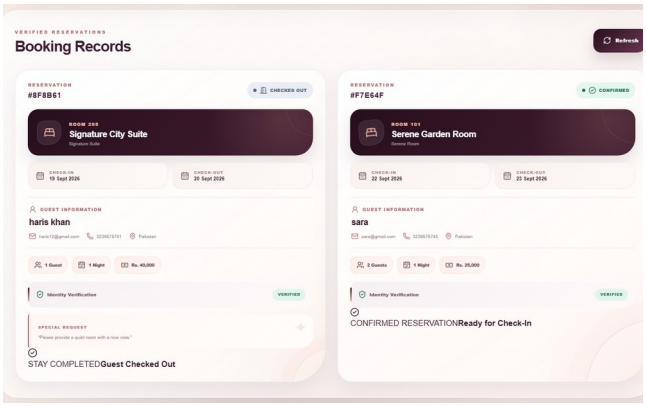
Feedback records include overall rating and category scores such as cleanliness, service and room experience. Reviewed feedback is visible to administrators for service-quality monitoring.

10. Manager Operations

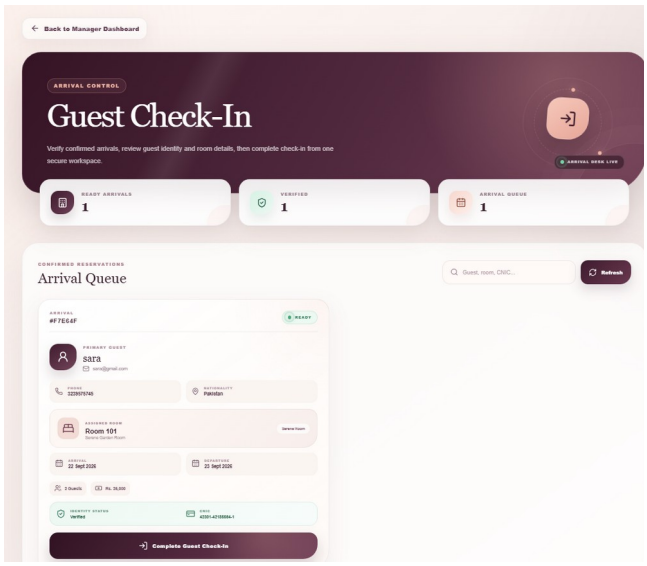
The Manager workspace handles day-to-day front-desk and operational activity.



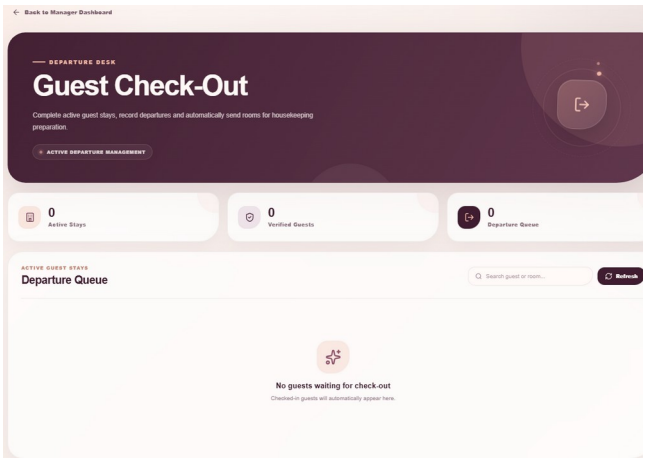
Manager Dashboard



Verified booking records

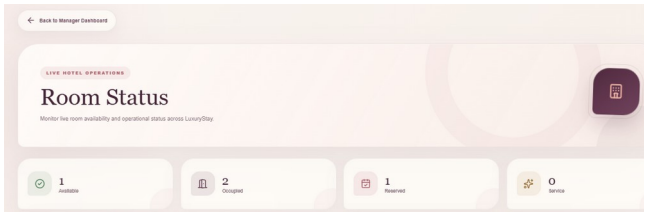


Guest Check-In

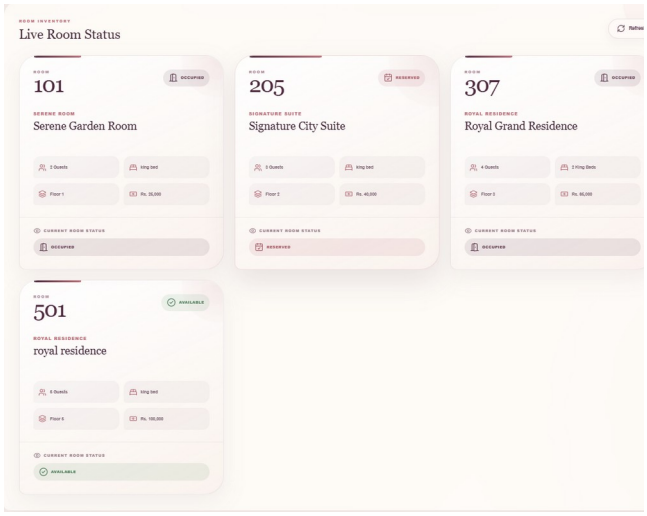


Guest Check-Out

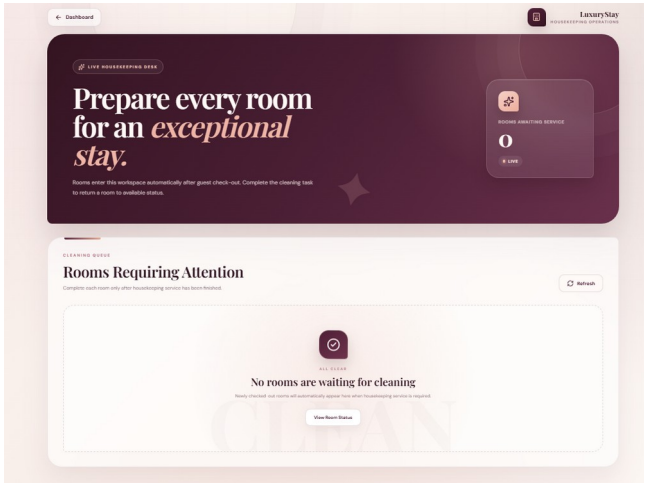
10.1 Rooms and Service Operations



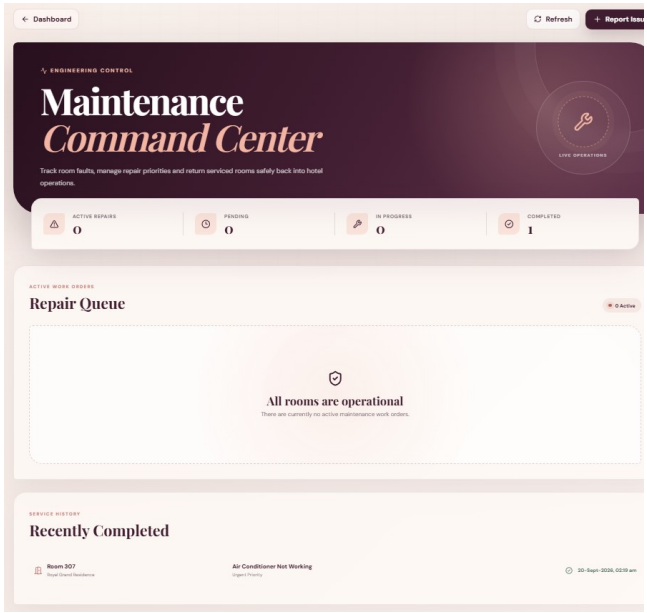
Room Status summary



Individual room operational states

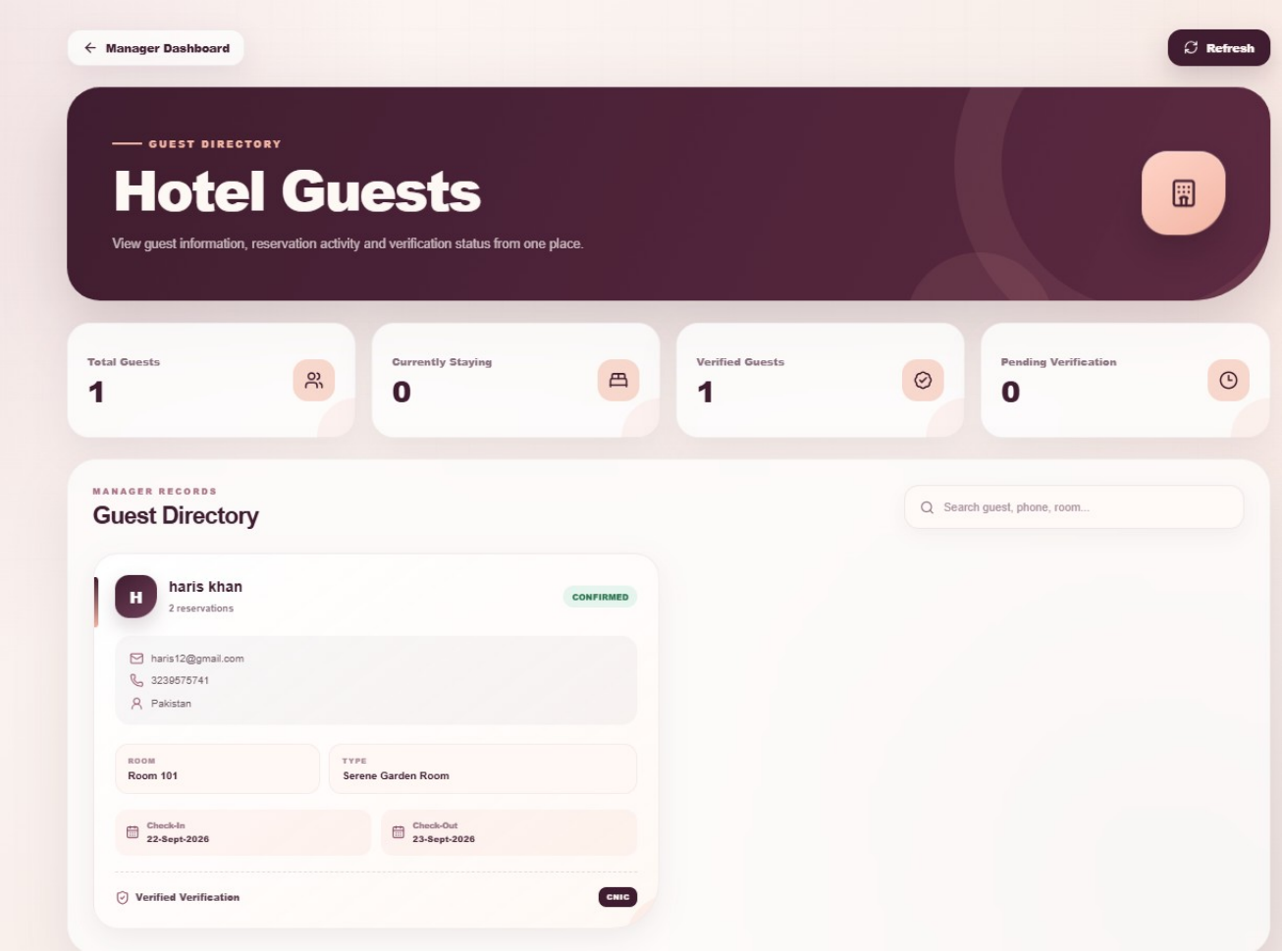


Housekeeping queue



Maintenance command center

10.2 Guest Directory

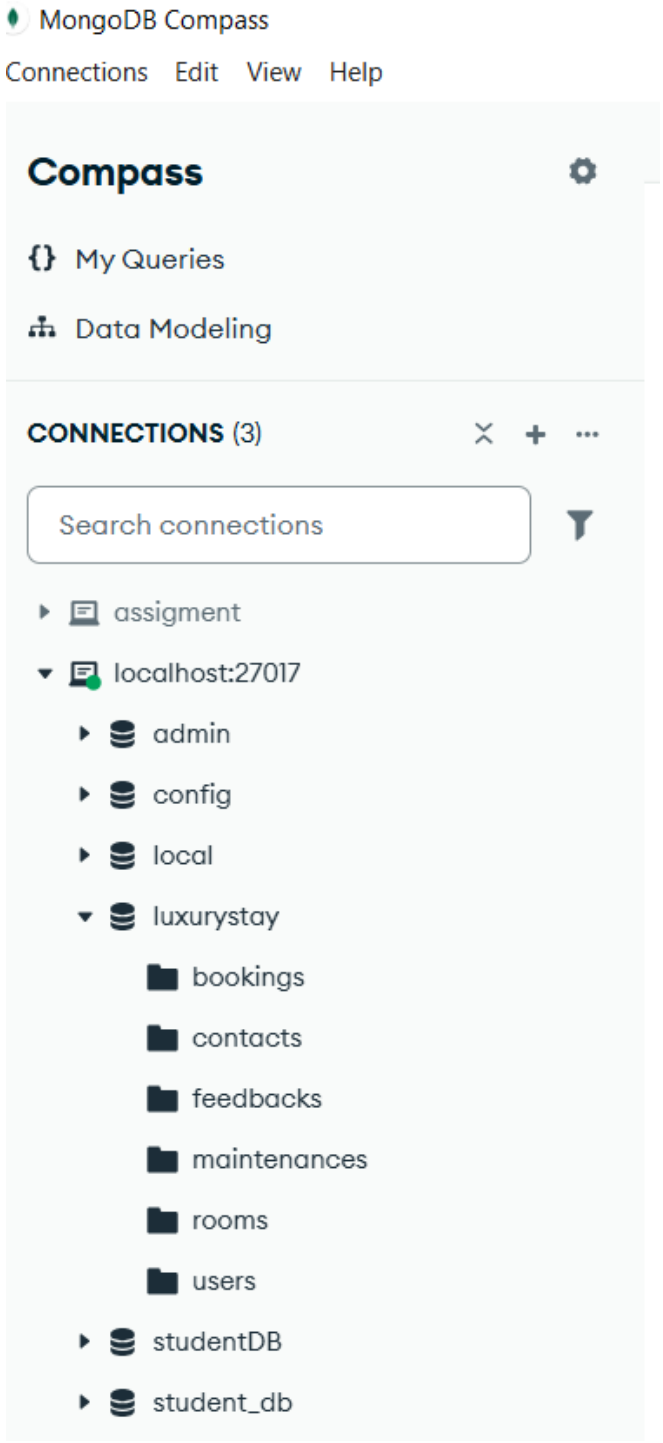


Guest Directory with reservation and verification information

- Confirmed arrivals appear in the check-in workflow.
- Active stays are processed through the departure workflow.
- Room status provides live operational visibility.
- Checked-out rooms can be prepared through housekeeping.
- Maintenance records track issues and completed work.
- Guest Directory consolidates reservation and verification information.

11. Database Design and Backend

11.1 MongoDB Collections

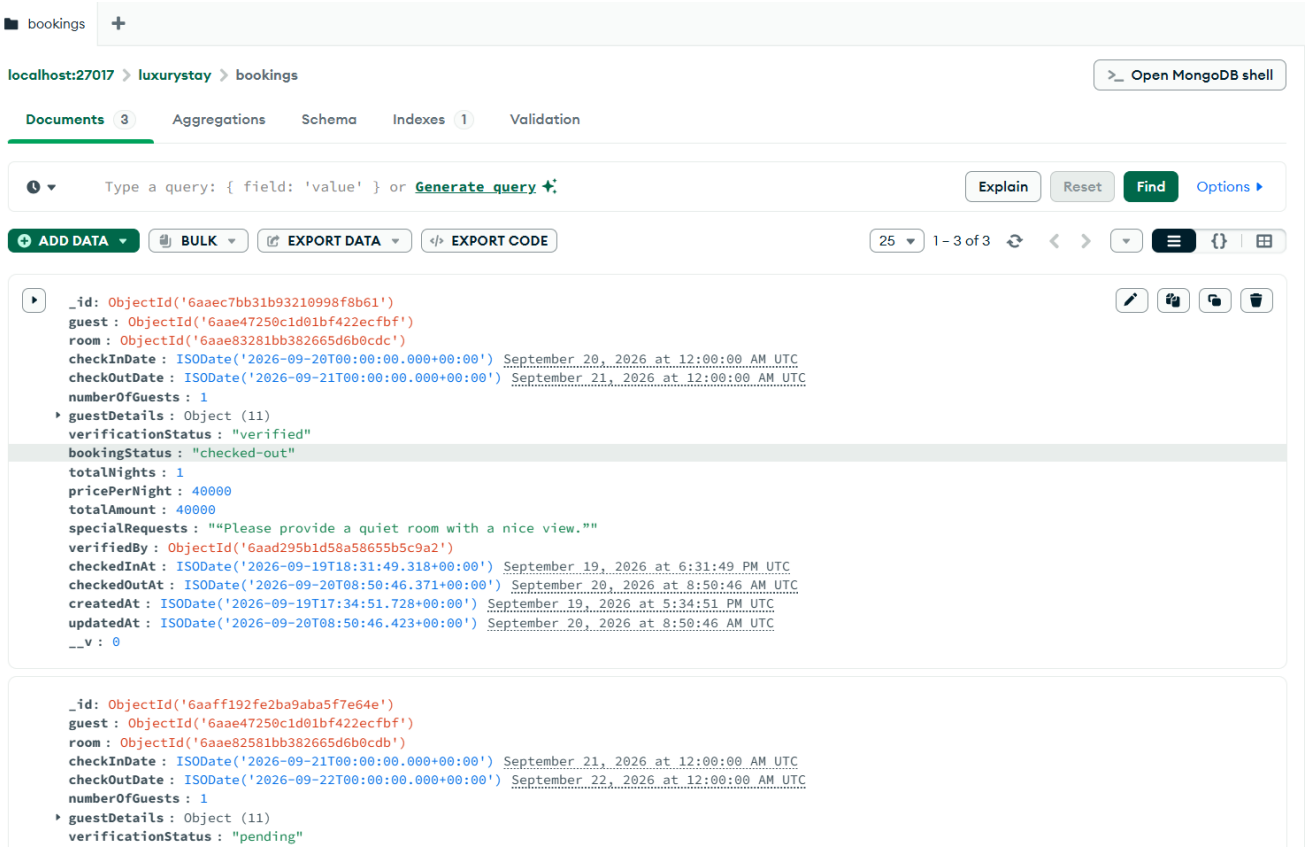


MongoDB Compass - application database and collections

Collection	Purpose
users	Guest, administrator and manager account records.
rooms	Room number, name, type, capacity, price and status.

bookings	Guest-room relationship, dates, amounts, verification and booking status.
contacts	Public contact/concierge enquiries.
feedbacks	Guest ratings, comments and review status.
maintenances	Maintenance issues, priority/status and completion history.

11.2 Booking Document Example



MongoDB booking document showing reservation fields

11.3 Backend Responsibilities

- Expose REST-style API routes for frontend modules.
- Validate requests and apply role-based authorization.
- Read and update MongoDB documents through Mongoose models.
- Maintain consistent booking, room and operational status transitions.
- Support identity document uploads with controlled file types and size limits.

12. Security and Validation

- Protected application areas require authenticated access.
- Role-based routing separates Guest, Administrator and Manager functions.
- Passwords are handled through the authentication layer rather than displayed in the interface.
- Booking verification and identity information are restricted to operational workflows.
- Uploaded identity files are limited to supported formats and configured file-size limits.
- Sensitive environment values should remain in environment configuration and must not be included in the submitted source package.

Final Submission Security Check

Before packaging the source code, regenerate any development JWT secret that may have been exposed during testing, exclude .env files from the ZIP, remove test credentials from source code and ensure uploaded identity documents are not included unless specifically required for demonstration.

13. Testing and Results

Test Scenario	Expected Result	Result
Guest registration/sign in	Correct role opens protected workspace.	Pass
Room availability	Available rooms and prices load correctly.	Pass
Reservation request	Booking is stored and visible to administration.	Pass
Admin confirmation	Confirmed booking becomes operationally available.	Pass
Guest check-in	Stay is activated and room becomes occupied.	Pass
Guest check-out	Stay completes and room can move to service workflow.	Pass
Housekeeping	Completed cleaning returns room to available status.	Pass
Maintenance	Issue can be tracked and completion retained in history.	Pass
Finance/Reports	Live booking and payment data is summarized.	Pass
Feedback	Completed-stay feedback is stored and visible to admin.	Pass
MongoDB persistence	Core records remain available after refresh.	Pass

Result Summary

End-to-end manual testing confirmed the main project flow from reservation creation through administrative processing and hotel operations. The screenshots in this report document the working interfaces and representative database records.

14. Limitations and Future Enhancements

14.1 Current Limitations

- Payment tracking is implemented as hotel billing/collection management rather than a production third-party payment gateway.
- Automated email and SMS notifications are not part of the demonstrated implementation.
- Advanced forecasting, dynamic pricing and enterprise-scale load testing are outside the current project scope.
- Automated cloud backup and disaster-recovery orchestration are not demonstrated.

14.2 Future Enhancements

- Integrate a secure online payment gateway and downloadable invoices.
- Add email/SMS booking confirmations and operational alerts.
- Introduce demand forecasting and dynamic room pricing.
- Add configurable taxes, hotel policies and seasonal rate rules.
- Implement automated backup, audit logging and production monitoring.
- Expand automated testing and accessibility validation.

15. Conclusion

LuxuryStay Hotel Management System demonstrates a complete role-based hotel workflow using the MERN stack. The project connects guest reservations with administrative control and manager operations while maintaining persistent hotel data in MongoDB.

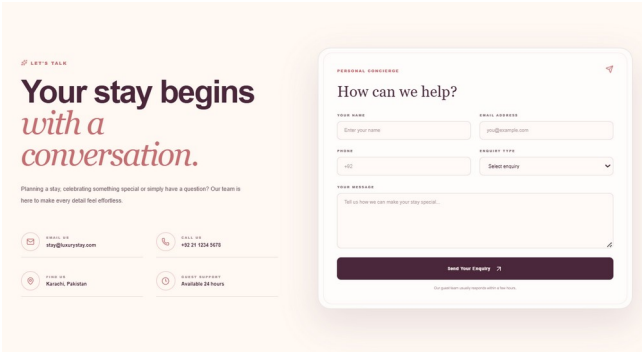
The final system provides a practical foundation for a real hotel-management product: it is modular, visually consistent and capable of supporting the key reservation, room, guest, service, finance and reporting processes required by the project scope.

Demonstration Sequence

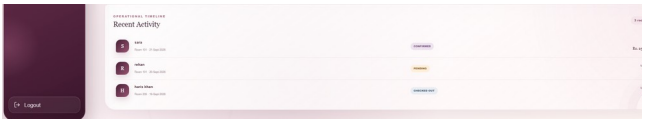
- Public website and room discovery
- Guest sign in / registration and reservation request
- Admin booking confirmation and room management
- Manager check-in, room status, check-out and housekeeping
- Finance and reporting
- Feedback and MongoDB records

Appendix A. Interface Screens

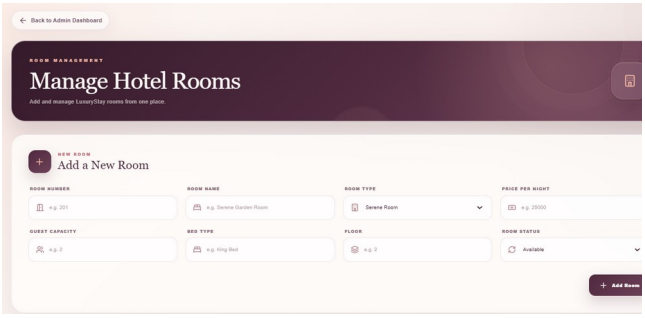
Additional interface evidence is included here without repeating module explanations.



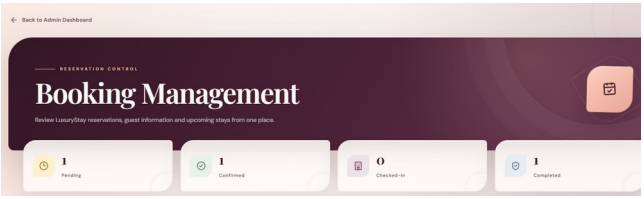
Contact / Personal Concierge



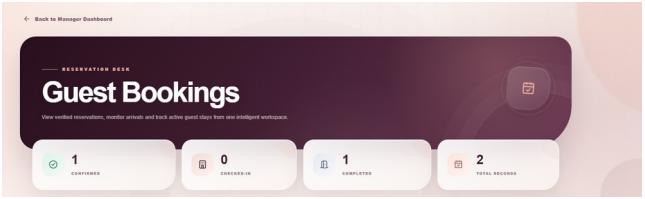
Admin recent activity and logout



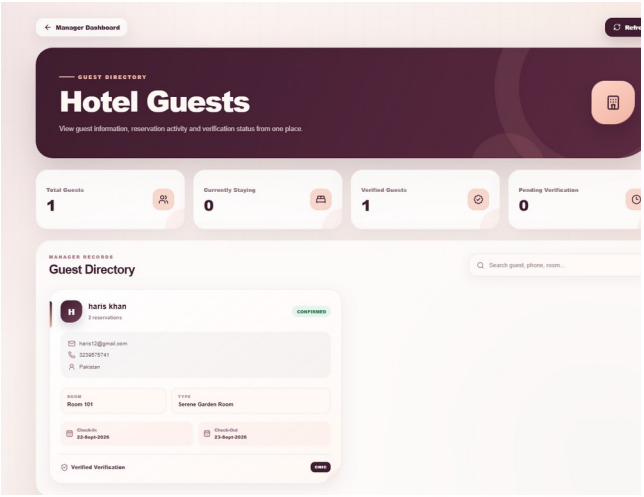
Add Room form



Booking Management summary



Manager booking summary



Guest Directory